

Mojia Zhang

+86 13856071128 | 567 West Yangsi Road, Shanghai, China | mz4441@nyu.edu

EDUCATION

NYU Shanghai

Computer Science

- Cumulative GPA: 4.0

Sep '25 - Present

Shanghai, China

EXPERIENCE

Machine Learning Force Field (MLFF) for Molecular Simulation

Research Experience | AI for Science

Mar '26 - Present

Shanghai, China

- Worked with the **GemNet neural network model (PyTorch)** to learn atomic forces and energies for molecular systems
- Processed molecular datasets (e.g., **MD17 aspirin**) and prepared training/validation/test data. Trained models on an HPC cluster (GPU), configuring training scripts and parameters (e.g., training steps, batch size). Monitored training progress using logs and Weights & Biases (offline mode), interpreting metrics such as **loss, energy MAE, and force RMSE**.
- Gained hands-on experience with a typical ML research pipeline.

TREE – a Pharmaceutical Decision Support AI Agent

The Second Place, NYU Shanghai Digital Innovation Competition, core technical builder

Dec '25 - Feb '26

Shanghai, China

- Developed an AI agent platform for pharmaceutical decision support, leveraging **multi-agent architecture** and **GraphRAG** to assist end-to-end drug development analysis for Henlius Biotech.
- Contributed to backend development using Python **FastAPI**, building API routes for data retrieval, LLM orchestration, and integration with external sources such as PubMed, ClinicalTrials.gov, and FDA.

Vibe Coding Hackathon, NYU Shanghai & NYU Abu Dhabi

Awarded "Embracing the Exponential" Prize

Sep '25 - Oct '25

Shanghai, China

- Guided by **AI collaborators** (Claude, Cursor, ChatGPT), rapidly learned and applied **full-stack development** from scratch to build an intelligent meeting spot recommender web application, mastering core concepts in Python Flask backend APIs and JavaScript frontend interaction.
- Translated conceptual prompts into technical solutions by integrating the Baidu Maps API and implementing a geometric calculation algorithm, navigating challenges such as CORS and system integration through iterative debugging.

LEADERSHIP ACTIVITIES

NYU Shanghai Digital Innovation Community

DIC Leadership Trainee Program (LTP)

Sep '25 - Present

Shanghai, China

- Assisted in planning and executing a **cross-university innovation competition**, including defining competition themes, designing rules and schedules, and managing participant registration and communication, resulting in **over 50 participating teams**.
- Supported the external liaison team by helping to draft invitation materials and coordinate communication with professors and industry partners like Tencent and Henlius, contributing to the successful engagement of experts and professors as judges or guest speakers.

AWARDS

- Certificate of Distinction**, Euclid Mathematics Contest 2024 — University of Waterloo

SKILLS

- Computer Skills: Python
- NLP Foundations: Familiar with core NLP architectures, including CNN/RNN models, attention mechanisms, word embeddings, and Transformer-based frameworks for language modeling.
- Mathematics: Foundational Calculus, Probability and Statistics
- Languages: Chinese and English (IELTS 7.0)
- Tools & Systems: Linux, Git